Med2Story

A Domain-Specific Extension of Text2Story for Clinical Narratives

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Preface

The production and organisation of temporal and referential information in clinical narratives is essential for advancing the management and analysis of health data. The ability to identify, annotate, and structure temporal and referential elements in clinical texts not only promotes diagnostic and therapeutic accuracy but also facilitates medical research, the development of computational systems, and interoperability across technological platforms. It was with this motivation that the present manual was created, aimed at guiding professionals, researchers, and students in the complex task of annotation in clinical texts.

This manual describes a temporal and referential annotation scheme that combines theoretical and practical approaches, with a specific integration of knowledge from the medical field. The scheme presented here was developed on Text2Story annotation framework (Silvano 2021, 2023, 2024; Leal 2022) and the standards established by the international ISO 24617-1:2012 norm: Language resource management—Semantic annotation framework (SemAF)—Part 1: Time and events (SemAF-Time, ISO-TimeML), which ensure uniformity and consistency in temporal annotation processes, and ISO 24617-9:2019 norm: Referential Annotation Framework, for the annotation of participants.

A.2. Participants

A.2.1. Definition:

Named entities, or participants, that play an important role in the event or state.

A.2.2. Tag span:

(i) **Noun Phrases** (*Doente de 73 anos; sistema nervoso central*)

(ii) **Pronouns** (*eles*; *estas*; a*quilo; o; lhe; que*)

(iii) **Adverbs** (*onde*)

(iv) **Adjectives** (*lombar*; *hepático*; *mesentéricas*)

Note 1:

Markables in the referential layer correspond primarily to noun phrases (NPs), including determiners and modifiers (1–2). When a full NP contains multiple referential units, the complete NP should be annotated, and smaller embedded NPs should also be annotated separately. For example, in the expression *"um tumor no pulmão esquerdo"*, the following markables should be identified: *"um tumor no pulmão esquerdo"*, *"um tumor"*, and *"o pulmão esquerdo"*.

When an NP includes quantificational information (e.g., numerals or quantity expressions), this information should be included in the markable span in this layer (1).

Clausal modifiers (including infinitival, gerundive, participial, and relative clauses) must be excluded from NP markables and annotated as separate markables (3).

Markables may also correspond to adverbial phrases (ADVPs) when they denote referential information, such as in (4). Finally, adjectives may be annotated as markables when they convey information related specifically to anatomical location (5).

(1) *Uma avaliação por TC confirmou a existência de uma lesão d*<participant>*a amígdala de 3cm*<participant> *associada a* <participant>*uma adenopatia de 17mm*<participant> *a nível d*<participant>*o hilo renal*<participant>*.*

(2) *Presença de* <participant>*células epiteliais com morfologia em anel de sinete*<participant>.

(3) *Proeminência d*<participant>*a região da apófise uniforme do pâncreas*<participant> *com indefinição d*<participant>*a primeira porção do duodeno*<participant>*, associando-se a um moderado espessamento difuso d*<participant>*a parede do antro gástrico*<participant> *com dificuldade de esvaziamento d*<participant>*o estômago*<participant>*.*

(4) *Foi transferida para o IPO,* <participant>*onde*<participant> *foi confirmado o diagnóstico.*

(5) <participant>*Doente de 35 anos*<participant>*, sem antecedentes relevantes, com uma história de uma cólica* <participant>*renal*<participant> *em fevereiro de 2017.*

Note 2:

In the case of appositive phrases denoting the age of an entity (typically a person), annotation must be performed using only one markable, regardless of whether a comma separates the noun phrase from the appositive.

(6) <participant>*Doente de 22 anos de idade*<participant> *sem antecedentes relevantes.*

(7) <participant>*Doente, de 22 anos de idade,*<participant> *sem antecedentes relevantes.*

A.2.3. Attributes

A.2.3.1. Lexical head

(i) **Noun**

(8) *Foi transferida para o IPO, onde foi confirmado o diagnóstico.*

(ii) **Pronoun**

(9) *Doente com história de múltiplas neoplasias, que se encontrava em tratamento de leucemia mieloide aguda*.

(iii) **Adverb**

(10) *Foi transferida para o IPO, onde foi confirmado o diagnóstico.*

(iv) **Adjective**

(11) *Fez uma avaliação medular com mielograma.*

A.2.3.2. Domain

A.2.3.2.1. Individuation

(i) **Set** – a set of individuals; it occurs only with collective nouns.

(12) *O conteúdo da consulta de grupo foi reportado à equipa médica assistente.*

(ii) **Individual** – a single individual.

(13) *Decide-se propor o doente para tratamento com Idarrubicina e Citarabina.*

(iii) **Mass** – a mass quantity.

(14) *Propõe-se alotransplante de medula óssea.*

A.2.3.3. Type

The *type* attribute is divided into eight general types (Figure 1), each of which is further subdivided into more specific *labels*.

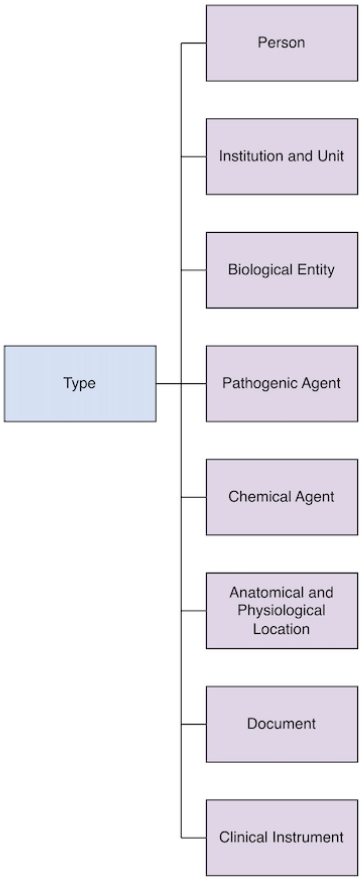

Figure 1 General Types of Participants

(i) **Person –** Entities referring to individual human beings involved in a clinical or healthcare context, including patients, healthcare professionals, caregivers, or other identified individuals. This label covers both explicitly named persons and role-based references (e.g., *paciente, doente, médico assistente, dador*). The attributes associated with Person are presented in Figure 2.

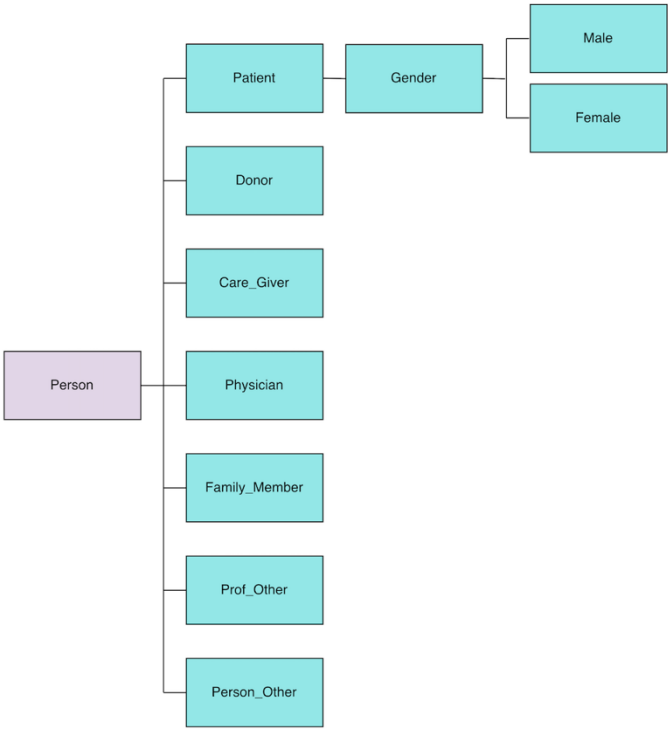

Figure 2 Attributes of Person

a. **Patient –** A person who receives, or is intended to receive, medical attention, care, or treatment, or who is registered with a healthcare professional or institution for the purpose of receiving medical care when required (e.g., *paciente, doente, doente do sexo feminino*). This label has two attributes: male and female. When it is not possible to identify the patient's sex from the context, no sex-related attribute is assigned.

(15) <participant, noun, individual, 1, person, patient, female>*Doente do sexo feminino com 29 anos*<participant> *e um quado recente caracterizado por astenia.*

b. **Donor –** Individuals supplying living tissue, organs, cells, blood or blood components for transfer or transplantation to histocompatible recipients (e.g., *dador*).

(16) *Decide-se propor o doente para alotransplante de medula óssea se* <participant, noun, individual, 1, person, donor>*dador HLA*<participant> *compatível.*

c. **Care_Giver –** Person who provide care to those who need supervision or assistance in ilness or disability (e.g., *cuidador*)

(17) *O doente foi acompanhado por* <participant, noun, individual, 1, person, care_giver>*um cuidador*<participant>*, que referiu dificuldades na gestão da medicação em contexto domiciliário.*

d. **Physician –** A doctor; a person who has been educated, trained, and licensed to practice the art and science of medicine (e.g., *médico, oncologista, hematologista, médico assistente*).

(18) *De acordo com* <participant, noun, individual, 1, person, physician>*o médico assistente do doente*[1]<participant>*, trata-se de um doente com critérios de vulnerabilidade.*

e. **Family_Member –** A person identified as having a familial relationship with the patient (e.g., *esposa, irmão, filho, marido*). This category includes explicit references to kinship used to contextualise family history, social support, or decision-making processes.

(19) *O doente tem* <participant, noun, individual, 1, person, family_member>*três irmãos compatíveis*<participant>*.*

f. **Prof_Other –** A person identified as a professional interacting with the patient in a clinical context who is not a physician. This category includes, for example, nurses, social workers, allied health professionals, therapists, psychologists, healthcare administrative staff, and other non-physician professionals involved in the patient's care, management, or follow-up.

(20) *O doente foi avaliado pel*<participant, noun, individual, 1, person, prof_other>*a enfermeira de traigem*<participant>*, que*[2] *procedeu à colheita de sinais.*

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[1] Within the syntagm *"o médico assistente do doente"*, *"o doente"* e "*o médico assistente"* would also be annotated with the attributes <participant, noun, individual, 1, person, patient, male> and <participant, noun, individual, 1, person, physician>

[2] The relative pronoun "*que*" should also be annotated with the same attributes as its referent, "*enfermeira de triagem*".

g. **Person_Other –** A person mentioned in the clinical text who is neither the patient, nor a family member, nor a clearly identifiable healthcare professional. This category includes generic or ambiguous references to individuals (e.g. *someone*, *another person*), as well as persons whose role is not clinically relevant or cannot be determined from the available context.

(21) *Foi trazido ao serviço por* <participant, noun, individual, 1, person, person_other>*uma pessoa conhecida*<participant>*, sem relação familiar identificada.*

(ii) **Institution and Unit –** Organisational entities involved in healthcare delivery, research, administration, or regulation, including hospitals, clinics, laboratories, healthcare centres, and their internal organisational units (e.g., wards, departments, services). The attributes associated with Institution and Unit are presented in Figure 3.

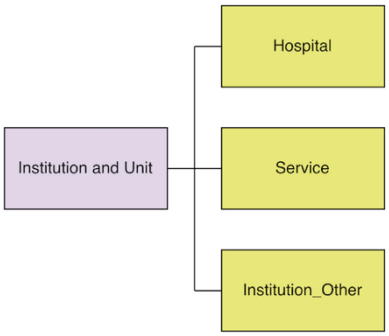

Figure 3 Attributes of Institution and Unit

a. **Hospital –** A healthcare institution that provides inpatient and/or outpatient medical care, including diagnosis, treatment, and monitoring of patients. This category includes general hospitals, specialised hospitals, and university or teaching hospitals.

(22) *Foi então transferida para* <participant, noun, individual, 1, Institution and Unit, Hospital>*o IPO*<participant>*, onde*[3] *foi confirmado o diagnóstico.*

b. **Service –** An organised clinical or healthcare unit, department, or functional service within a healthcare institution that delivers specific medical or support activities. This includes, for example, emergency services, oncology services, laboratory services, imaging services, and outpatient clinic.

(23) *Referenciação do doente para tratamento sintomático n*<participant, noun, individual, 1, institution and unit, service>*o Serviço de Medicina Paliativa do IPO Porto*[4]<participant>.

c. **Institution_Other –** An organisation or institution mentioned in the clinical text that is not a hospital and does not correspond to a specific clinical service. This category includes, for example, research institutions, universities, laboratories external to hospitals, healthcare authorities, insurance bodies, social care institutions, or other non-hospital organisations relevant to the patient's care or administrative context.

(24)*O relatório social foi posteriormente enviado para* <participant, noun, individual, 1, institution and unit, institution_other>*a Segurança Social*<participant> *para efeitos de avaliação de apoios*.

(iii) **Biological Entity –** Naturally occurring biological entities other than pathogenic agents, including cells, tissues, organs, genes, proteins, biological fluids (e.g., saliva, blood, semen), and other biological structures or substances relevant to human biology and medicine. The attributes associated with the Biological Entity are presented in Figure 4.

[3] The adverb "*onde*" should also be annotated with the same attributes as its referent, "*o IPO*".

[4] Within the syntagm *"o Serviço de Medicina Paliativa do IPO Porto"*, *o Serviço de Medicina Paliativa*" would also be annotated with the same attributes as "*o Serviço de Medicina Paliativa do IPO Porto*" and *"o IPO Porto"* would be annotated with the attributes <participant, noun, individual, 1, person, institution and unit, hospital>.

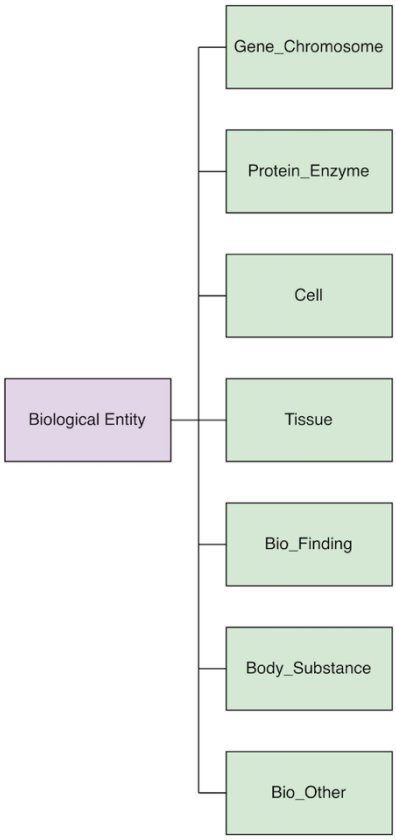

Figure 4 Attributes of Biological Entity

a. **Gene_Chromosome –** A molecular genetic entity corresponding to a gene, gene locus, or chromosome, including named genes, chromosomal regions, or mutations referenced in the clinical text (e.g., *o gene FLT3*, *FLT3*, (*16;16*), *NPM1*) .

(25) *Foi detetada a existência de uma mutação d* <participant, noun, individual, 1, biological entity, gene_chromosome>*o gene FLT3*<participant>*. No cariótipo foi identificada uma translocação d* <participant, noun, individual, 1, biological entity, gene_chromosome>*(16;16)*<participant>*.*

b. **Protein_Enzyme –** A biological macromolecule corresponding to a protein or enzyme, including structural proteins, receptors, enzymes, biomarkers, and other protein-based molecules involved in biological or pathological processes (e.g., *proteína, citólise, lípase, colestase, enzima*).

(26)*Analiticamente será de realçar uma hiperbilirrubinemia com aumento das* <participant, noun, individual, 1, biological_entity,

protein_enzyme>*enzimas de citólise e de colestase*[5]<participant> *e aumento da* <participant, noun, individual, 1, biological_entity, gene_chromosome>*lípase*<participant>.

c. **Cell –** A biological unit corresponding to a cell or specific cell type, including normal, abnormal, mature, or immature cells mentioned in the context of physiological or pathological states (e.g., *blastos, hemácias, leucócitos*).

(27) *O aspirado medular mostrou 48% de* <participant, noun, individual, >1, biological_entity, cell>*blastos de fenótipo monocitoide*<participant>.

d. **Tissue –** An aggregate of similar cells and their extracellular matrix forming a structural or functional unit within the body. This category includes references to tissues but excludes whole organs (e.g., *tecido ósseo, tecido cutâneo, tecidos moles)*.

(28) *A biópsia revelou infiltração neoplásica n*<participant, noun, mass, und, biological_entity, tissue>*o tecido ósseo*<participant>.

e. **Bio_Finding –** An observable or measurable biological, molecular, cellular, or histopathological finding resulting from clinical evaluation, laboratory testing, imaging, or diagnostic procedures. This includes abnormalities, markers, or states but excludes diseases or diagnoses (e.g., *uma massa, conglomerados adenopáticos, adenopatias)*

(29)*Os exames complementares revelaram* <participant, noun, individual, 1, biological_entity, bio_finding>*uma massa de partes moles*<participant>. *Apresentava, ainda,* <participant, noun, individual, >1, biological_entity, bio_finding>*conglomerados adenopáticos*<participant>.

f. **Body_Substance –** A natural substance produced by the human body and present in bodily fluids or secretions, including blood, plasma, serum, urine, cerebrospinal fluid, and other biological fluids (e.g., *sangue, plasma, esperma, urina*).

[5] Within the syntagm *"enzimas de citólise e de colestase"*, *"citólise" e "colestase"* would also be annotated with the attributes <participant, noun, individual, 1, person, biological_entity, protein_enzyme>.

(30) *Foi realizada recolha de* <participant, noun, mass, und, biological_entity, body_substance>*esperma*<participant> *para criopreservação.*

g. **Bio_Other –** Any biological entity mentioned in the clinical text that does not fall into the categories of Gene_Chromosome, Protein_Enzyme, Cell, Tissue, Bio_Finding, or Body_Substance. This includes biological concepts or entities that are relevant but insufficiently specified or not covered by the predefined biological categories.

(31) *Observou-se a presença de* <participant, noun, mass, und, biological_entity, body_substance>*material biológico*<participant> *não especificado na amostra analisada.*

(iv) **Pathogenic Agent –** Biological entities capable of causing disease or infection in humans, such as bacteria, viruses, fungi, parasites, or other infectious agents. The attributes associated with Pathogenic Agent are presented in Figure 5.

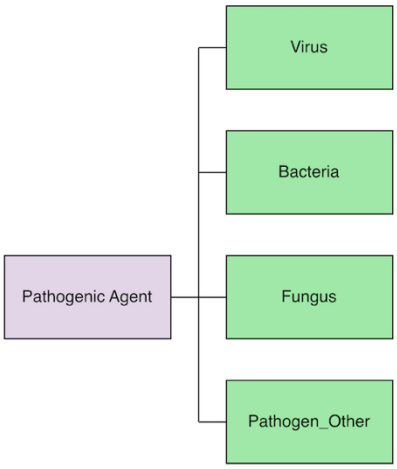

Figure 5 Attributes of Pathogenic Agent

a. **Virus** (e.g., *gripe A, H1N1, corona virus*)

(32)*O teste PCR foi positivo para* <participant, noun, individual, 1, pathogenic_agent, virus>*o vírus da gripe A*<participant>.

b. **Bacteria** (e.g., *Klebsiella pneumoniae, S. aureus MSRA*)

(33) *A urocultura identificou* <participant, noun, individual, 1, pathogenic_agent, bacteria>*Klebsiella pneumoniae*<participant>*, sensível à cefalosporina de terceira geração*.

c. **Fungus** (e..g, *Candida albicans, Cryptococcus neoformans*)

(34) *Infeção por* <participant, noun, individual, 1, pathogenic_agent, fungus>*Candida albicans*<participant>*.*

d. **Pathogen_Other –** An infectious biological agent mentioned in the clinical text that does not fall under the categories of Virus, Bacteria, or Fungus, or that is referred to in a generic or unspecified manner. This includes parasites, prions, or non-specified pathogens.

(35) *Suspeita-se de infeção por* <participant, noun, individual, 1, pathogenic_agent, pathogen_other>*agente patogénico* <participant> *não especificado, aguardando resultados laboratoriais adicionais.*

(v) **Chemical Agent –** Substances with a defined chemical composition that are relevant to biological or medical contexts, including drugs, pharmacological substances, laboratory reagents, contrast agents, and other chemical compounds used for diagnostic, therapeutic, preventive, or research purposes. The attributes associated with Chemical Agent are presented in Figure 6.

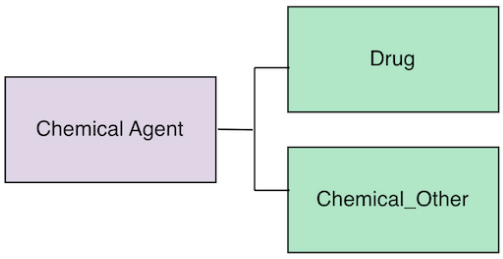

Figure 6 Attributes of Chemical Agent

a. **Drug –** A chemical substance used for the prevention, diagnosis, treatment, or management of disease or medical conditions. This category includes prescription drugs, over-the-counter medications, vaccines, and therapeutic agents administered to the patient (e.g., *Citarabina, Idarrubicina, Tazobac, Vancomicina*).

(36) *Decide-se propor o doente para tratamento com* <participant, noun, mass, und, chemical_agent, drug>*Idarrubicina*<participant> *e* <participant, noun, mass, und, chemical_agent, drug>*Citarabina*<participant>.

b. **Chemical_Other –** A chemical substance mentioned in the clinical text that is not used as a drug and does not fall under the Drug category. This includes reagents, contrast agents, disinfectants, solvents, toxins, or other chemical compounds relevant to clinical, laboratory, or environmental contexts.

(37) *Foi administrado* <participant, noun, mass, und, chemical_agent, chemical other>*contraste iodado*<participant> *antes da realização da TC.*

(vi) **Anatomical and Physiological Location –** Entities denoting anatomical structures, body parts, organs, body regions, or physiological spaces, as well as locations defined in functional or physiological terms (e.g., *vascular system*, *respiratory tract*). The attributes associated with Anatomical and Physiological Location are presented in Figure 7.

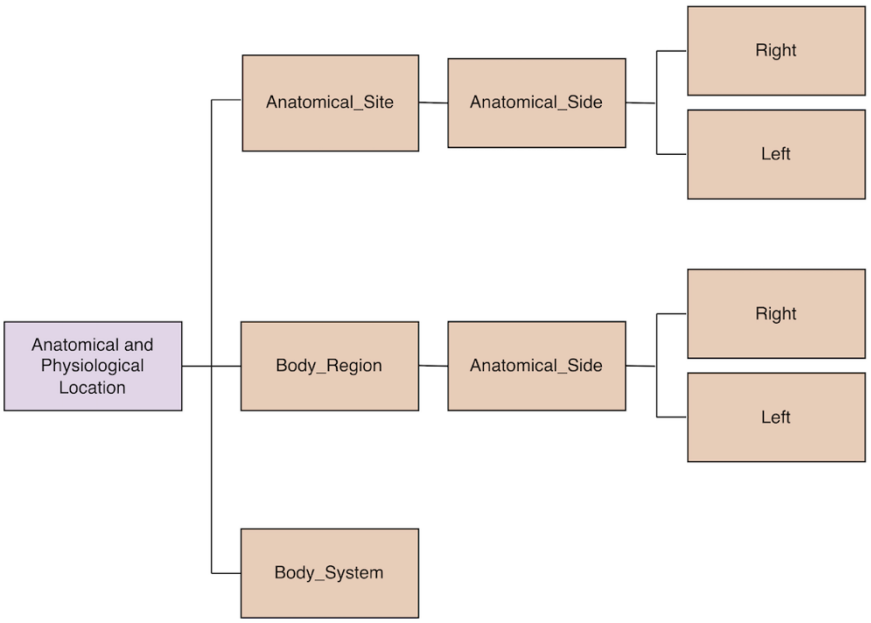

Figure 7 Attributes of Anatomical and Physiological Location

a. **Anatomical_Site –** A specific internal anatomical organ or structure within the human body where a clinical condition, procedure, finding, or intervention occurs. This category includes organs, anatomical structures, cavities, or precise sites (e.g., *colon, pelvis, bladder wall, mesorectal fascia, bone marrow*). This category may be further specified by the *Anatomical_Side* attribute, which indicates laterality and is classified as *Right* or *Left*.

(38) *Associadamente, refere-se diagnóstico de TVP d*<participant, noun, individual, 1, anatomical and physiological location, anatomical_site>*a veia gemelar do membro inferior direito*[6]<participant>*, tendo sido hipocoaguada.*

b. **Body_Region –** Anatomical areas of the body (e.g., *upper limb*, *lower limb*, *ankle*) (cf. (38).

c. **Body_System –** Anatomical structure, each instance of which has as its direct parts instances of one or more organ types which are interconnected with one another by zones of continuity.

(39) *Quadro clínico e analítico compatível com o diagnóstico de uma leucemia mieloide aguda com envolvimento do Sistema Nervoso Central.*

(vii) **Document –** Information-bearing artefacts that record, store, or communicate clinical, administrative, or scientific information, such as clinical reports, medical records, discharge summaries, consent forms, guidelines, or correspondence.

(40) <participant, noun, individual, 1, document, report>*O relatório social*<participant> *foi posteriormente enviado para a Segurança Social para efeitos de avaliação de apoios*.

(viii) **Clinical Instrument –** Medical devices, tools, or equipment used in clinical practice for diagnostic, therapeutic, monitoring, or procedural purposes, including both invasive and non-invasive instruments, as well as clinical scales, scores, and standardised assessment or classification systems (e.g., performance status scales such as ECOG,

[6] Within the syntagm *"a veia gemelar do membro inferior direito"*, *"o membro inferior direito"* would also be annotated with the attributes <participant, noun, individual, 1, anatomical and physilogical location, body_region, anatomical_side, right>.

disease staging systems). The attributes associated with Clinical Instrument are presented in Figure 9.

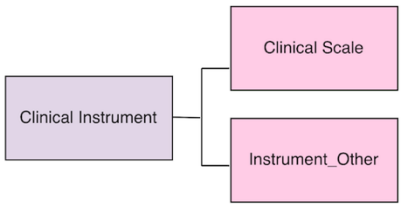

Figure 9 Attributes of Clinical Instrument

a. **Clinical Scale –** A standardised clinical assessment tool or scoring system used to measure, quantify, or classify a patient's symptoms, functional status, disease severity, or clinical outcomes. This category includes named scales, scores, indices, and rating systems used in clinical practice (e.g., *ECOG, estadio, IPI*).

   (40) *Apresentava um* <participant, noun, individual, >1, clinical_instrument, clinical_scale>*ECOG*<participant> *de 4. Trata-se de um linfoma não Hodgkin difuso de grandes células B,* <participant, noun, individual, >1, clinical_instrument, clinical_scale>*estadio*<participant> *IV,* <participant, noun, individual, >1, clinical_instrument, clinical_scale>*IPI*<participant> *de 5.*

b. **Instrument_Other –** A medical or clinical instrument, device, or tool mentioned in the clinical text that is not a clinical scale. This category includes diagnostic, monitoring, therapeutic, or procedural instruments and devices used in patient care.

   (41) *Foi monitorizado com* <participant, noun, individual, 1, clinical_instrument, instrument_other>*um oxímetro de pulso*[7]<participant> *durante o procedimento.*

[7] Within the syntagm *"um oxímero de pulso"*, *"pulso"* would also be annotated with the attributes <participant, noun, individual, 1, anatomical and physiological location, body_region>.

A.2.3.3. Involvement

(i) 0 – the empty set.

(42) *Não houve* <participant, noun, individual, 1, person, donor>*nenhum dador HLA compatível*<participant>.

(ii) 1 – a set with only one entity.

(43) *Decide-se propor* <participant, noun, individual, 1, person, patient>*o doente*<participant> *para tratamento.*

(iii) >1 – a set with more than one entity, but less than the totality of entities in the domain.

(44) *O aspirado medular mostrou 48% de* <participant, noun, individual, >1, biological_entity, cell>*blastos de fenótipo monocitoide*<participant>.

(iv) ALL - the totality of entities in the domain (= universal quantification).

(45) <participant, noun, individual, all, biological_entity, cell>Todas as células<participant> tinham anomalidas.

(v) UND - undefined involvement.

(46) *A biópsia revelou infiltração neoplásica n*<participant, noun, mass, und, biological_entity, tissue>*o tecido ósseo*<participant>.

A.2.4. Events vs. Participants

In Med2Story, events, i.e., eventive entities that denote something that occurs (e.g., "the patient underwent a full blood count.), are annotated in the temporal layer, whereas participants, i.e., entities with referential status that play an important role in the narrative (e.g., "the full blood count was added to the patient's electronic record"), are annotated in the referential layer. However, distinguishing between eventive and referential readings is not always straightforward, requiring the application of linguistic tests that allow ambiguous contexts to be disambiguated in a systematic manner.

The tests we consider valid for distinguishing events from participants (eventive vs. referential readings) are the following:

(i) Compatibility with *durou* ["lasted"] + *x tempo* ["for x time"]);

Events are compatible with *durou* ["lasted"] + *x tempo* ["for x time"] (47), but participants are not (48).

(47) *O hemograma durou cinco minutos.*

[*"The blood count lasted five minutes."]*

(48) *O Sistema Nervoso Central durou cinco minutos.*

["*The Central Nervous System lasted five minutes."]

(ii) Compatibility with the progressive + perception verb *assistir* ["to watch"];

Events are compatible with progressive + perception verb *assistir* ["to attend"] (49), but participants are not (50).

(49) *O médico está a assistir ao exame.*

["The doctor is attending the exam."]

(50) *O médico está a assistir ao Sistema Nervoso Central.

["*The doctor is attending the Central Nervous System."]

(iii) Behaviour with *de*-phrases and *por*-phrases;

Events allow a *por*-phrase, but not two *de*-phrases (51), and participants allow two *de*-phrases, but reject the *por*-phrase (52).

(51) *O médico está a assistir ao exame do doente pelo radiologista/*do radiologista.*

["The doctor is attending the examination of the patient by the doctor/*of the radiologist."]

(52) *Tens aqui o exame do doente do radiologista/do radiologista do doente/*pelo radiologista.*

["You have here the exam of the patient of the radiologist/of the radiologist of the patient/*by the radiologist."]

(iv) Compatibility with occurrence verbs (*ocorrer*, *realizar-se* ["take place"]).

Events are compatible with occurrence verbs (*ocorrer*, *realizar-se* ["take place"]) (53), but participants are not (54).

(53) *O exame ocorreu/realizou-se às 15h00.*

["The examination took place at 3:00 p.m."]

(54) **O Sistema Nervoso Central ocorreu/realizou-se às 15h00.*

["*The Central Nervous System took place at 3:00 p.m."]

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